

RES.6-012 Introduction to Probability Lec 24, 25, 26
Textbook Problem Sets - Sections 7.1 to 7.4

Section 7.1 Problem Sets

1) $S = \{0, 1, 2, 3\}$

Memo Walkthrough:

~~Let $X_n = 0$~~

~~Let $X_n = i$ if $X_{n-i} = \text{last arrival}$ $i = \{1, 2, 3\}$~~

~~$X_n = 0$~~

~~$p_{00} = 0.2$~~

~~$p_{01} = 0.8$~~

$P(T=k | T \geq k)$

$p_{20} = P(T=3 | T \geq 3)$

~~Let $X_n = i$~~ $\rightarrow$ last arrival at time $n-i$
 $X_n = 0 \rightarrow$ an arrival just happened.

p_{00} (Prob arrival after arrival) = 0.2

$\therefore p_{01} = 0.8$

$(1 - 0.2)$

Always: $P_{x0} + P_{x(x+1)} = 1$ either happens or it does not.
 $p_{12} = 1 - p_{10}$

$p_{10} = 0$ as $P(k) \rightarrow P(2) = 0$

$\therefore p_{12} = 1 - 0 = 1$

~~p_{20}~~ ~~p_{21}~~

$p_{12} \rightarrow p_{23}$ or p_{20}

$$\begin{aligned} p_{20} &= P(T=3 | T \geq 3) = \frac{P(T=3)}{P(T \geq 3)} = \frac{0.3}{P(T=3) + P(T=4)} = \frac{0.3}{0.3 + 0.5} = \frac{0.3}{0.8} = \frac{3}{8} \end{aligned}$$

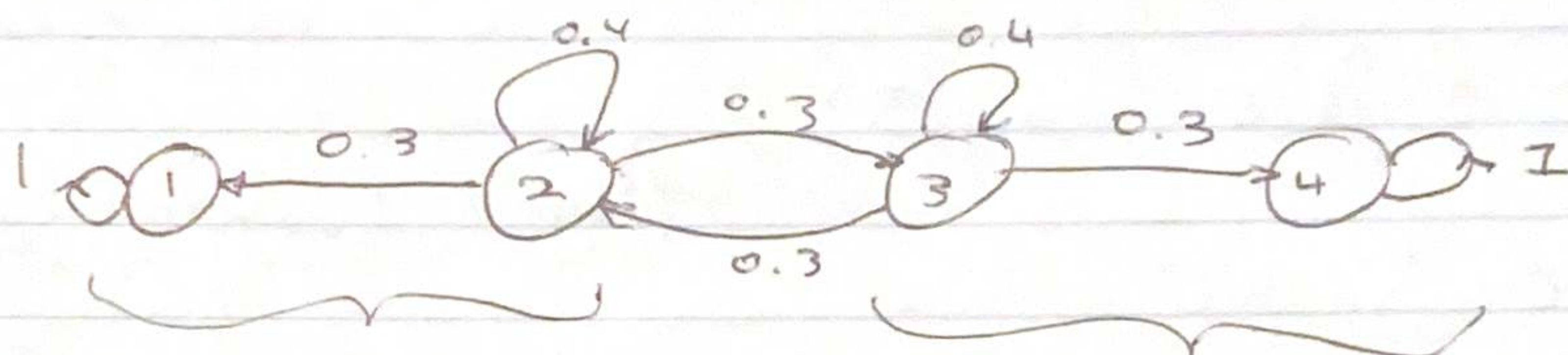
$\therefore p_{23} = 1 - 3/8 = 5/8$

~~p_{03}~~ $p_{03} = P(T=4 | T \geq 4) = \frac{0.5}{0.5} = 1$

3) $p(\text{move left}) = 0.3$ $p(\text{stay in place}) = 0.4$
 $p(\text{move right}) = 0.3$

Two spiders at positions 1 and m .

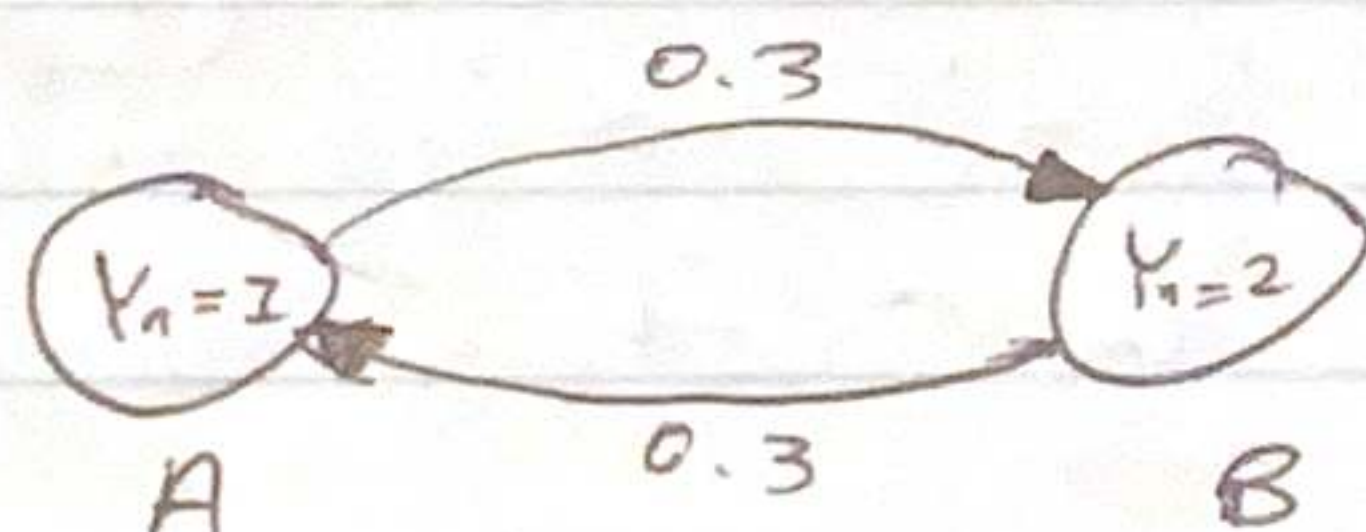
If Fly land on 1 or m , eaten.



Start in any state.

$$Y_n = 1 \quad \text{at state 1 or 2}$$

$$Y_n = 2 \quad \text{at state 3 or 4.}$$



Is Markov chain? $\rightarrow$ Yes, next state only depends on previous state.

No

$$\begin{matrix} P_{AA} \\ P_{AB} \\ P_{BB} \\ P_{BA} \end{matrix} =$$

Eventually, process trapped in S_1 or S_4
 so $p_{ij} \leq 0$ ($= 0$) for large enough number of switches

$$Y_n = \begin{cases} 1, & X_n \in \{1, 2\} \\ 2, & X_n \in \{3, 4\} \end{cases}$$

~~(Y_0, Y_1)~~ $\rightarrow$ Disprove Markov by showing Y_2 is dependent on Y_0 .

3 con.) To prove Y_2 depends on Y_0

$(Y_0, Y_1) = (1, 2)$ scenario 1
 $(Y_0, Y_1) = (2, 2)$ scenario 2
 }
 if Y_2 differs for these then Markov Property is broken.

Scenario 1:

$$(Y_0, Y_1) = (1, 2)$$

implies ① $X_0 = 2$ or 1 (if 1 X_1 cannot enable $Y_1 = 2$)
 so $X_0 = 2$

② $X_1 = 2$ or 3

but we know $Y_1 = 2$ (thus need $X_1 = 3$ or 4)

$\therefore X_1 = 3$

$$\begin{aligned}
 \therefore P(Y_2 = 2 \mid Y_0 = 1, Y_1 = 2) &= P(X_2 \in \{3, 4\} \mid X_1 = 3) \\
 &= 0.4 \{3\} + 0.3 \{4\} \\
 &= \underline{0.7}
 \end{aligned}$$

Scenario 2:

$$(Y_0, Y_1) = (2, 2)$$

implies ~~$X_0 = 2$~~ $X_0, X_1 \in \{3, 4\}$

$\therefore X_0 \in \{3, 4\} \therefore P(X_1 = 4 \mid Y_0 = Y_1 = 2) > 0$

$\hookrightarrow X_1$ from ~~3~~ 3 or 4 has pos probs

$$P(Y_2 = 2 \mid Y_0 = Y_1 = 2) \stackrel{\text{always}}{\longrightarrow} \text{trapped in } S = 4$$

$$\begin{aligned}
 &= P(Y_2 = 2 \mid X_1 = 4, Y_0 = Y_1 = 2)P(X_1 = 4 \mid Y_0 = Y_1 = 2) + \\
 &\quad P(Y_2 = 2 \mid X_1 = 3, Y_0 = Y_1 = 2)P(X_1 = 3 \mid Y_0 = Y_1 = 2) \\
 &= (1)[P(\dots)] + (0.7)[1 - P(\dots)]
 \end{aligned}$$

$$= 0.7 + (1 - 0.7)P(\dots)$$

$$= 0.7 + 0.3P(\dots) \quad \hookrightarrow \text{Do not need to know } \Rightarrow \underline{\underline{> 0}}$$

$$\neq 0.7 + (0.3)(p > 0)$$

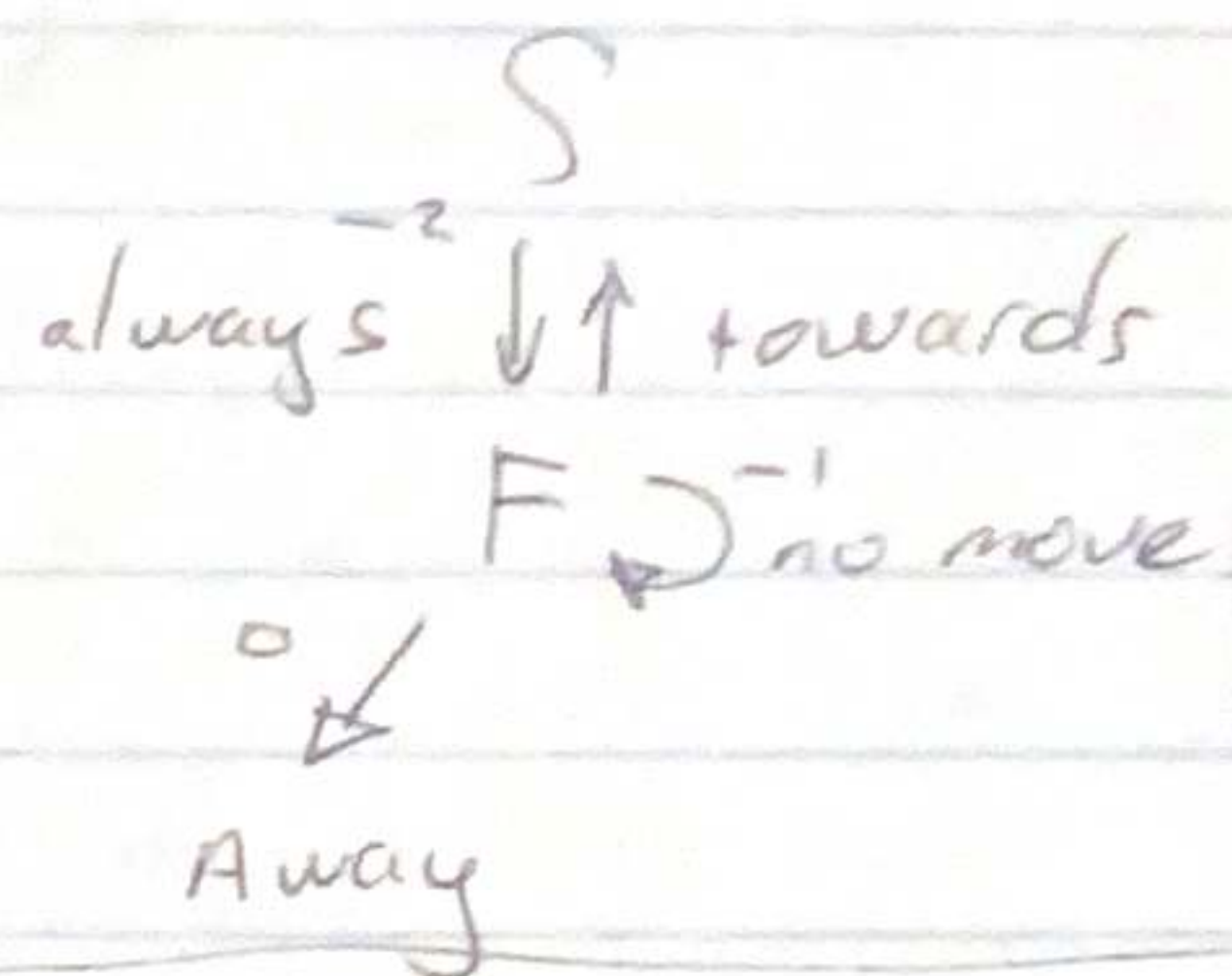
$$\neq > 0.7$$

Scenarios $\therefore P_{S1} < P_{S2} \therefore$ Markov prop fails

Section 7.2 Problem Set

4) $S \rightarrow$ move toward F always
 $F \rightarrow$ move toward S

- ① $p_{fs} = 0.3$
- ② $p_a = 0.3$
- ③ $p_{nm} = 0.4$



a) 3 possibilities each event, distance d : $\begin{cases} d+0 & \text{②} \\ d-1 & \text{③} \\ d-2 & \text{①} \end{cases}$
 $\therefore S = \{0, 1, \dots, d\}$

$$p_{00} = 1 \quad p_{0i} = 0 \text{ for } i \neq 0$$

$$p_{i(i-2)} = 0.3$$

$$p_{i(i-1)} = 0.4 \quad p_{ii} = 0.3 \text{ for } i \neq 0, 1$$

$$p_{ij} = 0 \text{ for } j \notin \{i-2, i-1, i\} \text{ otherwise}$$

$$p_{11} = \frac{0.3}{2} \rightarrow 0.15 \text{ ? mistake?}$$

$$p_{10} = \frac{0.4}{2} = 0.2 \text{ for } i \neq 0, 1$$

$$p_{11} = 0 \quad p_{10} = 0.4 + 0.3 = 0.7$$

b) Transient : States $d = \{1, \dots, n\}$
 Recurrent : State $d = \{0\}$

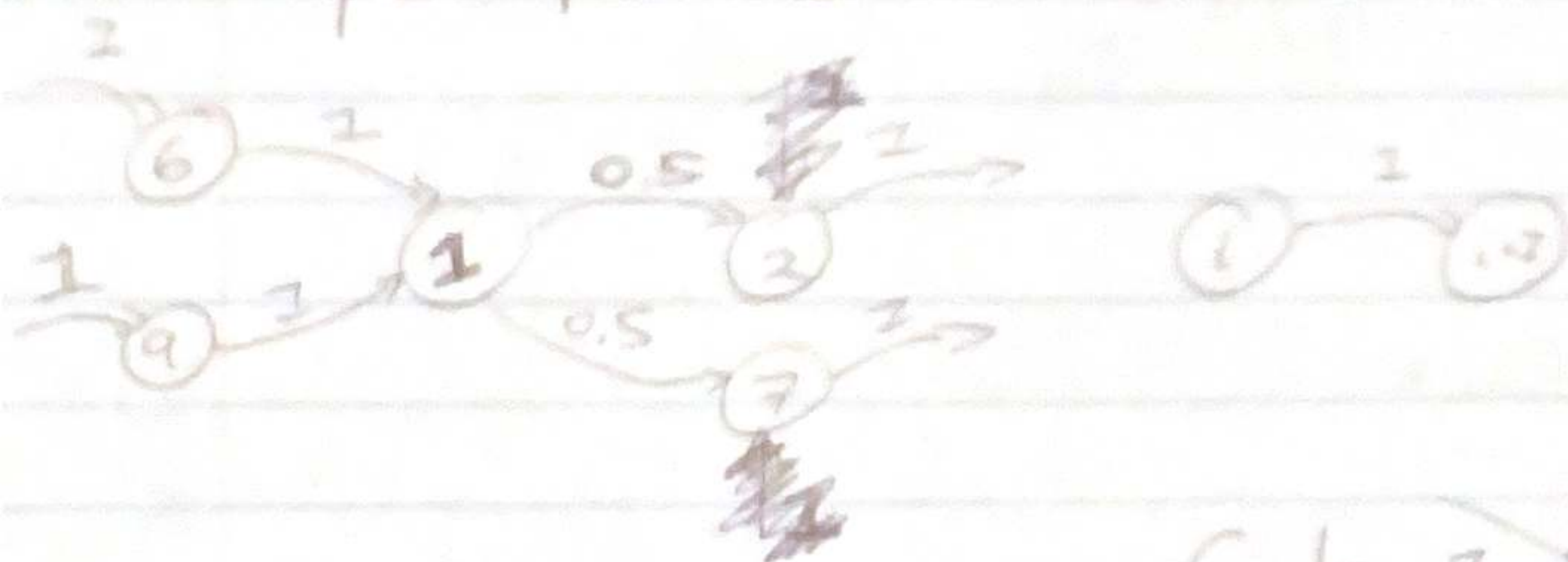
5)

$$S = \{1, 2, \dots, 9\}$$

$$p_{12} = p_{18} = \frac{1}{2}$$

$$p_{i(i+2)} = 1 \text{ for } i = 1, 6, 9$$

$$p_{41} = p_{91} = 1$$



$$\text{Cycle } 1: \{1, 2, 3, 4, 5, 6, 7\}$$

$$\text{Cycle } 2: \{2, 7, 8, 9, 1\}$$

$$\text{Periodicity} = 2 \checkmark$$

$$r_{ij}(n) > 0 \text{ for } i, j \in \{1, 2, \dots, 9\}$$

$$\begin{array}{l} \text{even } \{1, 3, 5, 8\} \\ \text{odd } \{2, 4, 6, 7, 9\} \end{array} \text{ steps from } (1)$$

Section 2.3 Problem Sets

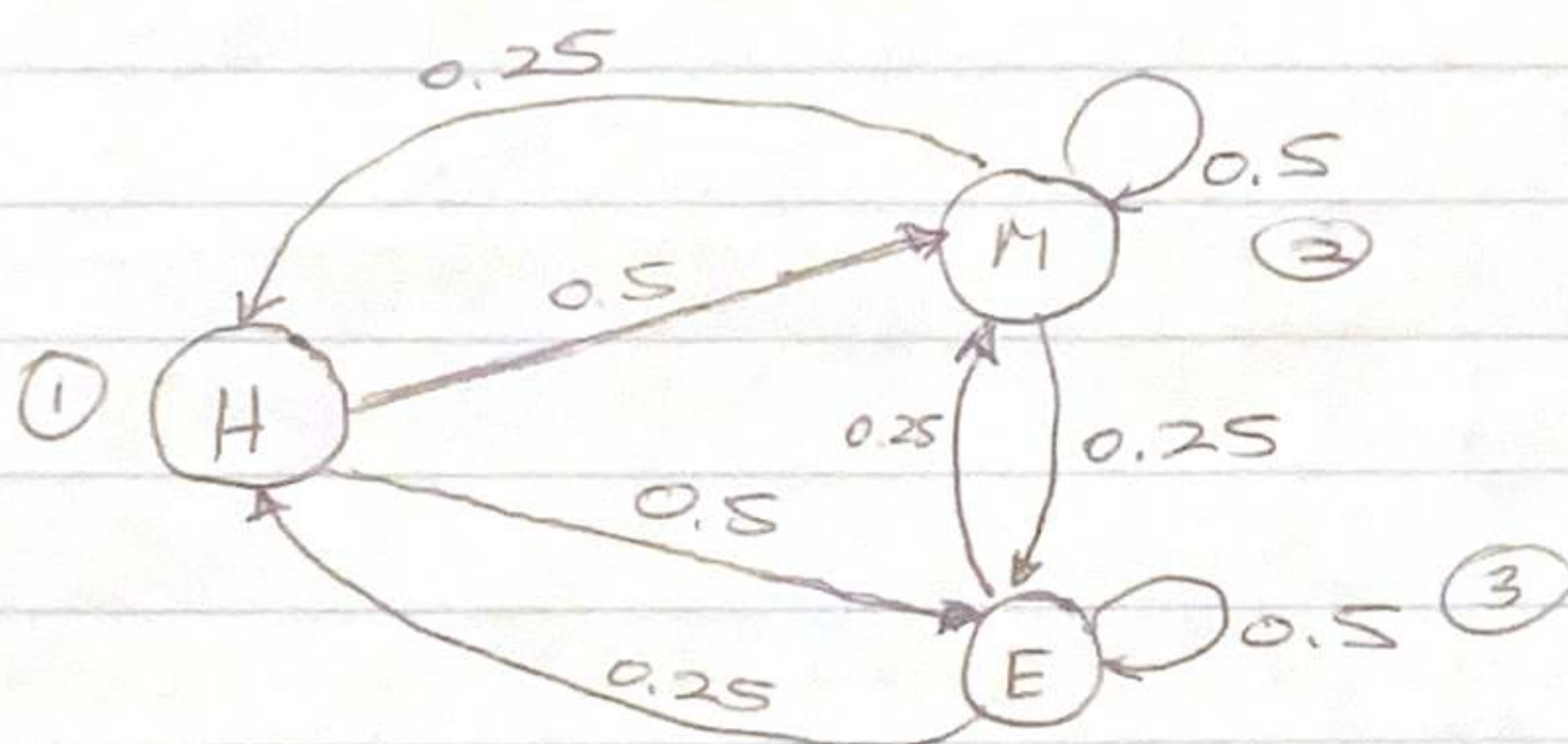
1140) Tests: hard - h, medium - m, easy - e

~~ti = h~~ $t_i = h$, then $t_{i+1} = m$ or e .

$t_i = e$ or m , then $t_{i+1} = 0.5(t_i) + 0.25(t_{i'}) + 0.25(t_{i''})$

$$S = \text{test } \{1, 2, 3, \dots, n\}$$

$$S = \text{test } \{h, m, e\} = \{1, 2, 3\}$$



$$P = \begin{matrix} & \begin{matrix} H & M & E \end{matrix} \\ \begin{matrix} H \\ M \\ E \end{matrix} & \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.25 & 0.5 & 0.25 \\ 0.25 & 0.25 & 0.5 \end{bmatrix} \end{matrix}$$

$$\pi_j = \sum_{k=1}^n \pi_k p_{kj}, \quad j = \{1, \dots, n\}$$

$$\sum_{k=1}^n \pi_k = 1$$

$$\pi_1 = \sum_{k=1}^n \pi_k p_{k1} = (0)\pi_1 + (0.5)\pi_2 + (0.5)\pi_3$$

$$= 0.5\pi_2 + 0.5\pi_3$$

$$= (0)\pi_1 + (0.25)\pi_2 + 0.25\pi_3$$

$$= \frac{1}{4}(\pi_2 + \pi_3) \quad \text{w prob to go } \pi_2 \rightarrow \pi_1$$

$$\pi_2 = (0.5)\pi_2 + (0.5)\pi_1 + 0.25(\pi_3)$$

$$\pi_2 = 2(0.5\pi_1 + 0.25\pi_3)$$

$$= \pi_1 + 0.5\pi_3$$

$$\pi_3 = (0.5)\pi_1 + (0.25)\pi_2 + (0.5)\pi_3$$

$$\pi_3 = \pi_1 + 0.5\pi_2$$

$$\therefore \pi_1 = \frac{1}{4} \left[\left(\pi_1 + \frac{1}{2} \pi_3 \right) + \left(\pi_1 + \frac{1}{2} \pi_2 \right) \right]$$

$$= \frac{1}{4} \pi_1 + \frac{1}{8} \pi_3 + \frac{1}{4} \pi_1 + \frac{1}{8} \pi_2$$

$$\frac{1}{2} \pi_1 = \frac{1}{8} (\pi_2 + \pi_3)$$

$$\frac{1}{2} \pi_1 = \frac{1}{8} (1 - \pi_1)$$

$$\frac{1}{2} \pi_1 = \frac{1}{8} - \frac{1}{8} \pi_1$$

$$\frac{5}{8} \pi_1 = \frac{1}{8}$$

$$\pi_1 = \frac{1}{5}$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$1 - \pi_1 = \pi_2 + \pi_3$$

$$\pi_2 = 1 - \pi_1 - \pi_3$$

$$\pi_3 = 1 - \pi_1 - \pi_2$$

$$\therefore \pi_2 = 1 - \left(\frac{1}{5} \right) - \pi_3 = \frac{4}{5} - \pi_3$$

$$\pi_3 = 1 - \left(\frac{1}{5} \right) - \pi_2 = \frac{4}{5} - \pi_2$$

$$\therefore \pi_2 = \pi_3 = \frac{2}{5} \checkmark$$

$$13) S = \{1, 2, 3\}$$

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} 0.4 & 0.6 & 0 \\ 0.3 & 0.5 & 0.2 \\ 0 & 0.2 & 0.8 \end{bmatrix}$$

$$a) \pi_1 = 0.4 \pi_1 + 0.6 \pi_2 + \cancel{0.2 \pi_3}$$

$$\pi_2 = 0.3 \pi_1 + 0.5 \pi_2 + 0.2 \pi_3$$

$$\pi_3 = 0.2 \pi_2 + 0.8 \pi_3$$

$$0.2 \pi_3 = 0.2 \pi_2$$

$$\pi_2 = \pi_3$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$\pi_1 = 1 - 2\pi_2$$

$$\cancel{0.4} \pi_1 = \cancel{0.4} \pi_2 = (1 - 2\pi_2)$$

$$3\pi_2 = 1$$

$$\pi_2 = \frac{1}{3} ; \pi_3 = \frac{1}{3}$$

$$\pi_1 = \pi_2 = \frac{1}{3}$$

$$13) \begin{bmatrix} \pi_{11} & \pi_{12} & \pi_{13} \\ \pi_{21} & \pi_{22} & \pi_{23} \\ \pi_{31} & \pi_{32} & \pi_{33} \end{bmatrix} = \begin{bmatrix} 0.4 & 0.6 & 0 \\ 0.3 & 0.5 & 0.2 \\ 0 & 0.2 & 0.8 \end{bmatrix}$$

$$\pi_1 = 0.4\pi_1 + 0.3\pi_2$$

$$0.6\pi_1 = 0.3\pi_2$$

$$2\pi_1 = \pi_2 \quad \pi_1 = \frac{1}{2}\pi_2$$

$$\pi_2 = 0.6\pi_1 + 0.5\pi_2 + 0.2\pi_3$$

$$= 0.8\pi_2 + 0.2\pi_3$$

$$0.2\pi_2 = 0.2\pi_3$$

$$\pi_2 = \pi_3$$

$$\text{so } \pi_1 = \frac{1}{2}\pi_3$$

$$\pi_3 \checkmark$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$\pi_1 + \frac{1}{2}\pi_1 + \pi_1 = 1$$

$$\frac{5}{2}\pi_1 = 1$$

$$\pi_1 = \frac{1}{5}$$

$$\pi_1 = \frac{1}{5} ; \quad \pi_2 = \pi_3 = \frac{2}{5} \checkmark$$

$$b) \quad b_i = p_{i,i+1} \quad b_i = p_{i,i+1}$$

$$P(\text{First observed transition is a birth}) = \sum_i \pi_i b_i$$

$$b_1 = p_{12} = 0.6$$

$$b_2 = p_{23} = 0.2$$

$$b_3 = 0$$

$$\begin{aligned} \sum_i \pi_i b_i &= \left(\frac{1}{5}\right)(0.6) + \left(\frac{2}{5}\right)(0.2) + \left(\frac{2}{5}\right)(0) \\ &= \frac{1}{5} \checkmark \end{aligned}$$

$$c) P(\text{Birth} \mid \text{change of state}) = \frac{\sum_i \pi_i b_i}{\sum_i \pi_i (b_i + d_i)}$$

$$\sum_i \pi_i b_i = \frac{1}{5}$$

$$d_1 = 0$$

$$d_2 = p_{21} = 0.3$$

$$d_3 = p_{32} = 0.2$$

$$\sum_i \pi_i (b_i + d_i) = \left(\frac{1}{5}\right)(0.6 + 0) + \left(\frac{2}{5}\right)(0.2 + 0.3) + \left(\frac{2}{5}\right)(0 + 0.2) = \frac{3}{5}$$

$$P(\text{birth} \mid \text{change of state}) = \frac{(\frac{1}{5})}{(\frac{3}{5})} = \frac{1}{3} \quad \times$$

Memo: $\sum_i \pi_i \frac{b_i}{b_i + d_i}$

$$\begin{aligned} &= \left(\frac{1}{5}\right)\left(\frac{0.6}{0.6+0}\right) + \left(\frac{2}{5}\right)\left(\frac{0.2}{0.2+0.3}\right) + \left(\frac{2}{5}\right)\left(\frac{0}{0+0.2}\right) \\ &= \left(\frac{1}{5}\right) + \left(\frac{2}{5}\right)\left(\frac{2}{5}\right) + \left(\frac{2}{5}\right)(0) \\ &= \frac{9}{25} \quad \checkmark \end{aligned}$$

$$d) P(\text{birth first transition} \mid \text{state 2 start}) =$$

~~$$\begin{aligned} &= \frac{P(\text{birth})}{P(\text{start state 2})} \\ &= \frac{b_2 \pi_2}{\pi_2} \\ &= \frac{(0.2)(\frac{2}{5})}{(\frac{2}{5})} \quad \times \end{aligned}$$~~

$$P(\text{state 2} \mid \text{first transition birth}) =$$

$$\begin{aligned} &= \frac{P(\text{state 2} \& \text{ birth})}{P(\text{birth first tran})} \\ &= \frac{(\frac{2}{5})(0.2)}{(\frac{9}{25})} = \frac{2}{9} \quad \checkmark \end{aligned}$$

$$\begin{aligned}
 e) P(\text{state 2} \mid \text{first change state birth}) &= \frac{P(\text{state 2 and first change birth})}{P(\text{first change birth})} \\
 &= \frac{\pi_2 \cdot \frac{b_2}{b_2 + d_2}}{\sum_i \pi_i \cdot \frac{b_i}{(b_i + d_i)}} \\
 &= \frac{\left(\frac{2}{5}\right) \left(\frac{0.2}{0.2 + 0.3}\right)}{\left(\frac{9}{25}\right)} \\
 &= \frac{4/25}{9/25} \\
 &= 4/9 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 f) P(\text{first transition birth} \mid \text{resulted in change of state}) &= \frac{P(\text{first transition birth and change of state})}{P(\text{change of state})} \\
 &= \frac{P(\text{birth})}{P(\text{change of state})} \\
 &= \frac{(1/5)}{\left(\pi_1 p_{12} + \pi_2 (p_{21} + p_{23}) + \pi_3 p_{32}\right)} \\
 &= \frac{(1/5)}{\left[\left(\frac{1}{5}\right)(0.6) + \left(\frac{2}{5}\right)(0.5) + \left(\frac{2}{5}\right)(0.2)\right]} \\
 &= \frac{(1/5)}{(2/5)} = \frac{1}{2} \quad \checkmark
 \end{aligned}$$

* $P(\text{steady state births}) = P(\text{steady state deaths})$

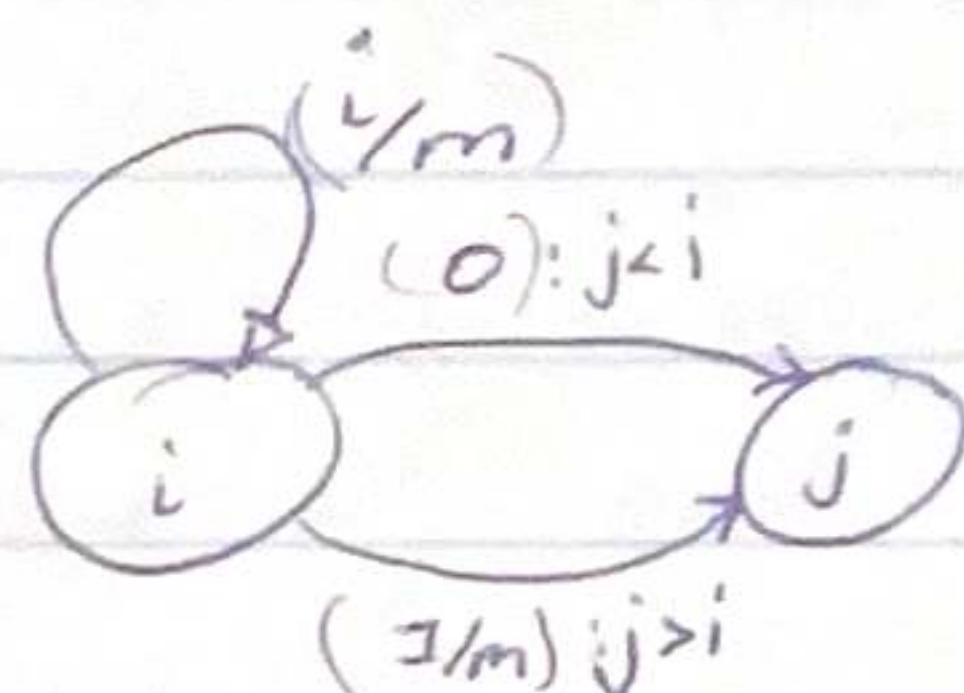
$$\begin{aligned}
 g) P(\text{First Trans to 2} \mid \text{change of state}) &= \frac{P(\text{First trans to 2 and change of state})}{P(\text{change of state})} \\
 &= \frac{\pi_1 p_{12} + \pi_3 p_{32}}{(2/5)} \\
 &= \frac{\left(\frac{1}{5}\right)(0.6) + \left(\frac{2}{5}\right)(0.2)}{(2/5)} \\
 &= \frac{1}{2} \quad \checkmark
 \end{aligned}$$

Section 7.4 Problem Sets

28) m classes rank 1 to m , m highest difficulty.
 $p = \frac{1}{m}$



$$\begin{cases} j < i : & p_{ij} = 0 \\ j > i : & p_{ij} = \frac{1}{m} \end{cases}$$



b) Recurrent state : $\{1\}$
 Transient state : $\{2, \dots, m\}$ $\times$ $\{m\}$
 $\{1, \dots, m-1\}$

c) $t_i = 1 + \sum_{j=1}^m p_{ij} \cdot t_j$ for $i \neq s$
 $= 1 + (\quad) \times$

→ Could use mean first passage time

→ Easier to view m as success within Bernoulli process

success prob = $\frac{1}{m}$

$E[t] = m$ years.

30 a) $R: \{6, 7\} \quad \{1, 2, 3\}$ ✓
 $T: \{4, 5\}$ ✓

class 1 $\{6, 7\} \rightarrow$ period 2. ✓
class 2 $\{1, 2, 3\} \rightarrow$ aperiodic ✓

b) $\begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.5 & 0.4 & 0.1 \\ 0 & 0.6 & 0.4 \end{bmatrix}$

$$\pi_1 = (0.5)\pi_1 + (0.5)\pi_2 + (0)\pi_3$$

$$0.5\pi_1 = 0.5\pi_2$$

$$\pi_1 = \pi_2$$

$$\pi_2 = 0.5\pi_1 + 0.4\pi_2 + 0.6\pi_3$$

$$\pi_2 = 0.9\pi_2 + 0.6\pi_3$$

$$0.1\pi_2 = 0.6\pi_3$$

$$\pi_1 = \pi_2 = 6\pi_3$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$2\pi_1 + \pi_3 = 1$$

$$\pi_1 = \frac{1}{2}(1 - \pi_3)$$

$$6\pi_3 = \frac{1}{2}(1 - \pi_3)$$

$$12\pi_3 = 1 - \pi_3$$

$$13\pi_3 = 1$$

$$\left\{ \pi_3 = \frac{1}{13} \quad \pi_1 = \frac{6}{13} \quad \pi_2 = \frac{6}{13} \right\}$$

Yes, steady states ✓

$$\pi_i = 0 \text{ for } i \notin \{1, 2, 3\}$$

$$\hookrightarrow \pi_i = 0 \text{ for } i \notin \{1, 2, 3\}$$

c) ~~$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$~~ $i \in \{6, 7\}$ given start in state 6.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\pi_6 = (0)\pi_6 + (1)\pi_7$$

$$\pi_7 = (1)\pi_6 + (0)\pi_7$$

$$\pi_6 = \pi_7$$

$$\pi_6 + \pi_7 = 1$$

$$\therefore \pi_6 = 0.5$$

$$\pi_7 = 0.5$$

$\rightarrow$ Periodic between states $\{6, 7\}$ so no

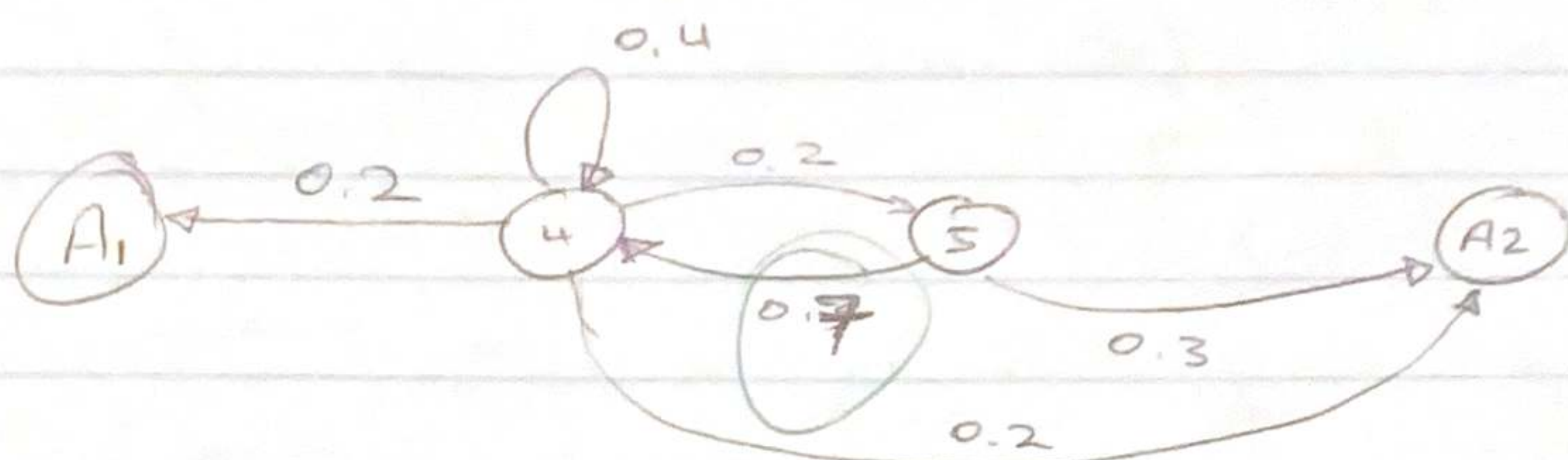
$$= \frac{1}{2} \quad \times$$

$$\hookrightarrow P\left(\frac{\text{state inc}}{\text{state change}}\right) = \frac{0.1}{0.1+0.5} = \frac{1}{6}$$

$$\begin{array}{r} 7 \\ 13 \\ \hline 20 \end{array}$$

e) Process start state 4

(i)



$$A_1 : \{1, 2, 3\}$$

$$A_2 : \{6, 7\}$$

$$a_i = \sum_{j=1}^n p_{ij} \cdot a_j$$

Let $a_{A_1} = a_1$
Let $a_{A_2} = a_2$

$$a_4 = (0)a_{A_1} + (0)a_{A_2} + (0.4)a_4 + 0.3(a_5) + (0.3)a_5$$

$$a_4 = 0.4a_4 + 0.3a_5$$

$$0.6a_4 = 0.3a_5$$

$$a_4 = \frac{1}{2}a_5$$

$$a_5 = 0.2a_4$$

$$a_4 = 0.2a_1 + 0.2a_2 + 0.2a_5 + 0.4a_4$$

$$0.6a_4 = 0.2(a_1 + a_2 + a_5)$$

$$0.6a_4 = 0.2(1 + 0 + a_5)$$

$$a_5 = 0.3a_4 + 0.3a_2$$

$$a_5 = 0.3a_4$$

set $a_1 = 1, a_2 = 0$

$$0.6a_4 = 0.2 + 0.2a_5$$

$$a_5 = 0.3a_4$$

set $a_1 = 0, a_2 = 1$

$$0.6a_4 = 0.2 + 0.2a_5$$

$$a_5 = 0.3a_4 + 0.3$$

$$0.6a_4 = 0.2 + 0.09a_4 + 0.09$$

$$0.6a_4 = 0.2 + 0.06a_4 + 0.06$$

$$0.54a_4 = 0.26$$

$$a_4 = \frac{13}{27}$$

$$\sum a_4 = 1?$$

Start State 4

$$P(A_1) = \frac{10}{27} \quad P(A_2) = \frac{13}{27}$$

$$a_4 = \frac{10}{27}$$

$$a_5 = \frac{1}{9}$$

30 e con.)

$$(ii) \mu_i = 1 + \sum_{j=1}^m p_{ij} \mu_j$$

$$\mu_4 = 1 + (0.4)\mu_4 + 0.2\mu_5$$

$$\mu_5 = 1 + 0.7\mu_4$$

$$\mu_4 = 1 + 0.4\mu_4 + 0.2 + 0.14\mu_4$$

$$0.46\mu_4 = 1.2$$

$$\mu_4 = \frac{1.2}{0.46} = \frac{60}{23} \checkmark$$